

# Alexander Yoza

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## Experience

- Capital One**, Associate Software Engineer – Richmond, VA February 2026 – Present
- Lead observability for a platinum-level availability service supporting up to **1B requests/day**, building dashboards, alerts, and health indicators to improve incident detection and operational readiness.
  - Developed **AWS CDK** infrastructure and **Go** backend services for high-availability service operations, improving deployment repeatability and service maintainability.
  - Built production alarms and service health indicators for latency, availability, and dependency failures, improving on-call visibility and incident response workflows.
  - Designed **AI-assisted engineering workflows** for debugging, documentation, test planning, and observability analysis while maintaining human code review and production-readiness standards.
- Nisatsu**, Founder & Software Engineer – Phoenix, AZ January 2026 – Present
- Designed and developed a mobile-first language-learning app with guided study flows, vocabulary practice, spaced review, and polished **multilingual UI/UX**.
  - Built scalable product architecture with **React Native**, **Next.js**, and **FastAPI**, translating learning goals into reusable components, clean data models, and AI-supported study flows.
  - Created **LLM-assisted product workflows** to accelerate product iteration, code review, content planning, and launch marketing while preserving structured review and product quality standards.
- Ponzu**, CTO & Co-Founder – San Diego, CA September 2025 – Present
- Led technical strategy and architecture for an **AI workflow orchestration platform**, defining product requirements, system design, and launch roadmap for an early-stage startup.
  - Built full-stack application architecture with **Next.js**, **TypeScript**, and **FastAPI**, supporting AI-powered workflow generation, structured task creation, and production-ready user flows.
  - Designed human-in-the-loop workflow patterns that convert ambiguous ideas into structured tasks, reusable prompts, and executable product plans.
- SitesByAlex**, Web Developer – Phoenix, AZ June 2023 – Present
- Delivered client websites and web applications using **Angular**, **Next.js**, **Svelte**, **TypeScript**, and **Tailwind CSS**, translating business requirements into responsive production-ready interfaces.
  - Designed, tested, and deployed responsive websites with emphasis on usability, cross-browser compatibility, performance, and maintainable client handoff.
- San Diego Supercomputer Center**, SDSC Developer Intern – San Diego, CA June 2024 – Aug 2024
- Led full-stack web development, guiding project direction, setting technical standards, and coordinating team execution in an agile environment.
  - Developed a dynamic web platform using **React.js**, **Node.js**, and **Firebase** to help students discover age-appropriate venues.
- Bank of Hawaii**, eSolutions Development Intern – Honolulu, HI June 2022 – Aug 2022
- Automated banking workflows with **Microsoft Power Automate**, saving over **1,500 hours annually**; developed **Python scripts** to process daily transactions and financial reports.

## Education

- University of California San Diego**, BS in Computer Science Sep 2021 – Dec 2024
- **Coursework:** Artificial Intelligence, Object-Oriented Programming, Data Structures, Software Engineering, Database Systems
- International Christian University**, Tokyo, Japan – Exchange Program Aug 2024 – Nov 2024
- **Japanese Language:** Conversational Level

## Projects

- Trading Lab** In Development
- Built a **Python** platform for researching, simulating, and scoring algorithmic trading strategies with typed strategy contracts, deterministic replay, strict plugin validation, and pre-fill risk checks.
  - Integrated support for **AI-generated strategies** by validating lifecycle hooks, enforcing import allowlists, applying schema validation, and running untrusted code in sandboxed **Docker** containers before execution.

## Technologies

**Languages:** TypeScript, Python, Go, Java, C, C++, HTML, CSS, JavaScript, SQL

**Technologies:** AWS, AWS CDK, FastAPI, React.js, Next.js, Node.js, Express.js, PostgreSQL, Prisma, MongoDB, Firebase, React Native, Docker, CI/CD, CloudWatch, Observability, Redis, Tailwind CSS, Angular, Svelte, Vercel, Git, GitHub, GitLab, Jest